

STATEMENT OF WORK

Nationwide A/E IDIQ for Dam Safety

Part I - Background

The Fish and Wildlife Service Coordination Act of 1934 established the statutory authority for the U.S. Fish and Wildlife Service (FWS) to operate facilities associated with Fish and Wildlife Conservation. Under this authority and several later statutes and executive orders, the FWS operates hydraulic structures that have been designed and constructed internally; acquired through purchase, use, and control; acquired through certain agreements; or use and control acquired through the Federal Aid program.

Presently, there are approximately 305 earthen embankment or concrete structures classified as dams (approximately 16 are considered High or Significant hazard) which are the responsibility of the FWS at various locations throughout the United States. These dams, levees, other hydraulic structures, and their appurtenant facilities are operated to promote conservation and enhancement of the fish and wildlife resource.

The Program is governed by the Federal Guidelines for Dam Safety, Department of Interior (DOI) Departmental Manual 753 chapters 1 & 2, and FWS Service Manual Chapter 361 FW 1-3 (Attachment 1,2,3), which describes the standards and requirements of the U.S. Fish and Wildlife Service's (Service) Dam Safety Program.

Part II - Objective

The objective of this contract is to provide comprehensive technical support for the dam safety program and for other civil engineering and water resource related responsibilities, including levees of the FWS. This contract may also provide comprehensive technical support for other bureau dam safety programs within DOI. The services expected to be performed under this contract may consist of, but are not limited to, inspection, risk analysis and Potential Failure Mode Analysis (PFMA), preparation and exercising of Emergency Action Plans (EAPs), engineering analyses, planning, design, construction supervision, repair, modification, new construction, and reconstruction of dams, levees, other hydraulic structures, their appurtenant facilities, and associated work.

Funded projects, that are not performed by in-house staff, will be accomplished through individual task orders, issued under this basic contract. Firm fixed-price task orders will include specific statements of work and be negotiated as individual projects.

Part III - Scope of Services and Responsibilities

The Architect/Engineer (A/E) shall furnish any or all of Safety Evaluation of Existing Dams (SEED) Inspections, Pre-Title I, Title I, Title II, Title III, and National Environmental Policy Act (NEPA) Compliance Services to the Government for projects at various locations throughout the United States. Typical projects may include, but are not limited to new dams, rehabilitation, reconstruction or modification of existing dams, levees, other hydraulic structures, their appurtenant facilities (such as spillways, outlet works, gates, etc.), and associated work.

Historically, dam inspections occur on a rotational schedule as required by each DOI bureau. However,

there are often unique, timely or special technical support or inspections that may require immediate attention. Task Orders are typically awarded each new year. Typical task orders may include, but are not limited to: inspections; analysis; feasibility; decommissioning; design; preparation of drawings, specifications, and other construction documents; field investigations; risk analysis and PFMA; stability studies; hydraulic and hydrologic studies; biological and cultural studies, documentation, construction supervision, consultation, and Clean Water Act certification and permitting.

The A/E is professionally responsible for the services performed under this contract. The A/E shall be responsive to Government-furnished criteria, information, guidance, and review comments. All services shall be in full compliance with Federal, State and bureau requirements.

Standard Responsibilities

1. The services specified in each task order could include visits to the project sites; meetings with Government representatives and other key personnel or consultants; and making presentations of proposed design solutions to selected officials. The location for these visits, meetings, and presentations will be specified in the individual task orders and could include the A/E's office, government offices, or any other site as deemed appropriate by the government for the particular requirement.
2. The A/E shall, without additional cost to the Government, obtain the necessary permits, licenses, and approvals from all local, State, and Federal authorities as are necessary for the performance of the services by the A/E required by the individual task orders issued under this contract. Should it become necessary in the performance of the work to secure the right of ingress and egress to perform any of the work on properties not owned or controlled by the Government, the A/E shall secure the consent of the representative, or agent prior to effecting entry on such property.
3. The A/E shall prepay shipping charges on all charts, sketches, drawings and documents submitted to the Project Manager, Contracting Officer (CO), or Contracting Officer's Representative (COR).
4. The A/E shall closely coordinate the work for which they are responsible under this contract with related work being performed by government personnel.
5. During the construction phase of a project, when required by the Government in a task order, the A/E shall check and recommend for approval/rejection submittals by the construction contractor (shop drawings, samples, equipment lists, data sheets, etc.) associated with the A/E's portion of the design work.
6. The A/E may be required to assist the Government in conferring and coordinating with state and local government agencies.
7. The A/E may be required to be a member of the final inspection team on all new construction and major repair and alteration projects; however, final acceptance shall be predicated exclusively upon the inspection performed by Government personnel.
8. The use of trade names, proprietary items, and the drafting of a specification by adapting a manufacturer's description of a commercial article is permissible but is not intended to be restrictive to free and open competition among suppliers. The item shall be specified by giving its functional or salient characteristics, such as chemical composition, physical characteristics,

laboratory test results, performance in actual use, etc. The phrase "or approved equal" shall be used when designating a commercial article or its salient characteristics.

Tasks to be Performed

As the Government identifies project requirements, a task order will be issued in accordance with the Task Order Procedures outlined in the award document. The A/E may be required to perform any or all of the following tasks:

1. SEED Inspections shall:

- a. Be conducted in accordance with the Federal Guidelines for Dam Safety (December 2023 FEMA P-93), Department of the Interior (DOI), bureau standards, publications, and accepted guiding engineering principles.
- b. Include an assessment of previous analyses.
- c. Identify problems and provide information so that the ~~Service~~ Government can make fully informed decisions regarding the need for repairs, operational changes, modifications, or additional analyses.
- d. Include a review of existing EAPs and Standing Operating Procedures (SOPs) of the high and significant hazard dams.
- e. Review the previous SEED reports.
- f. Perform an on-site examination to determine the current condition of the dam.
- g. Include a conference held between the inspecting engineer and the accompanying Government representative(s) proceeding and at the conclusion of the field inspection.
- h. Include an assessment of any changes in the floodplain and the adequacy of the current Hazard Classification.
- i. Include recommendations and cost estimates (Class IV) for additional risk, stability, seepage, hydrology- hydraulics, etc., analyses needed to assess the performance of the dam will be provided.
- j. Include recommendations and cost estimates to resolve identified deficiencies including operation, maintenance, monitoring, repair, or rehabilitation.

2. Pre-Title I (Project Identification/Planning/Feasibility Analysis) tasks could include any or all of the following:

- a. Determination of feasible engineering solutions/alternatives which satisfy the project goals, objectives, and user requirements.
- b. Studies to determine project scope and design criteria.
- c. Analysis of alternative sites.
- d. Risk Analysis.
- e. Dam Hazard Classification.
- f. Preparation of sketches.
- g. Development of a preliminary cost estimate range and schedule (Class IV cost estimate).
- h. Identification of applicable codes and design standards.
- i. Other task(s) as determined by project-specific requirements.

3. Title I (Conceptual Design) tasks could include any or all of the following:

- a. Collection and development of design data (e.g., surveying, drilling, sampling, testing, etc.).
- b. Performance of engineering analysis and studies (e.g., hydraulic, stability, seismic, etc.).
- c. Preparation of alternative conceptual designs.
- d. Preparation of a report presenting the various alternative designs including:

- 1) description of the study and investigations;
- 2) descriptions of the alternatives considered;
- 3) drawings and sketches required to illustrate the alternatives;
- 4) listing of data used and assumptions made;
- 5) evaluation of the alternative designs to include Class III cost estimates;
- 6) comparative evaluation of the alternative designs;
- 7) recommendation for final design, including discussion of the basis for the recommendation;
- 8) summary of results, conclusions, and recommendations.
- e. Performance of value engineering (VE) studies.
- f. Other task(s) as determined by project-specific requirements.

4. Title II (Final Design) tasks could include any or all of the following:

- a) Performance of design analysis and engineering calculations.
- b) Preparation of construction drawings.
- c) Determination and documentation of reservoir operation during construction.
- d) Determination and documentation of dewatering requirements and other construction considerations.
- e) Preparation of technical specifications.
- f) Preparation of a construction cost estimate (Class I cost estimate).
- g) Preparation of a material and equipment source listing and associated manufacturer's data.
- h) Documentation of designer's operating criteria.
- i) Other task(s) as determined by project-specific requirements.

5. Title III (Construction Management) tasks could include any or all of the following: Performance of construction inspection.

- a) Review of contractor submittals, Requests For Information (RFI).
- b) Preparation of a "first filling plan" as applicable.
- c) Preparation of construction progress reports.
- d) Photographing/documentation of construction progress.
- e) Review of VE change proposals submitted by the construction contractor.
- f) Review of construction contractor invoices, claims, modifications, etc.
- g) Development of record drawings.
- h) Completed Construction Engineering Report.
- i) Other task(s) as determined by project-specific requirements.

6. NEPA Compliance

The A/E may be required to perform studies, documentation, and consultations necessary to comply with the requirements of the NEPA and to obtain certification and permits required under the Clean Water Act. Tasks could include any or all of the following:

- a) Perform studies to determine impacts to wildlife, vegetation, water, soil, air, cultural resources, wetlands, floodplains, and recreation caused by the project design alternatives.
- b) Participate in Section 7 consultation for threatened and endangered species.
- c) Perform cultural resources activities in accordance with the requirements of the National Historic Preservation Act of 1966, including: conduct surveys, prepare reports, assess affects/impacts, and consult with the State Historic Preservation Officer (SHPO).
- d) Prepare NEPA documentation (Environmental Assessment, Environmental Impact Statement, Finding of No Significant Impact, Record of Decision, etc.)
- e) Conduct/participate in public involvement activities (scoping meetings, public hearings, etc.)

- f) Perform studies and prepare applications for certification and permits required under the Clean Water Act (i.e., Section 401, 402, 404).
- g) Develop design considerations for mitigation of potential impacts.
- h) Other task(s) as determined by project-specific requirements.

The A/E may be required to provide other design and/or construction data needed for state approvals, permits, etc. Specific requirements will be stated in individual task orders, as required.

Part IV - Requirements for Deliverables

Unless specified differently in a Task Order, the following requirements shall apply to submittals for Title I (Conceptual Design) and for Title II (Final Design). Submittal requirements for other levels of project development will be as designated in the individual task orders. All electronic source documents along with input and output files and a full pdf version of all submitted documents shall be required on a Government approved electronic file transfer site or device.

All data and information shall be provided in their current application to the Government.

- A.** Drawings shall be titled, numbered, and formatted to the standards used by the bureau, and shall be produced in AutoCAD (latest release) format. The 100-percent submittal package for final design shall include:
 - 1.** Copy of all AutoCAD files.
 - 2.** One copy of all input and out files from analytical software such as hydrology, hydraulics, slope stability, seepage, or stability.
- B.** Specifications shall comply with CSI (Construction Specifications Institute) format and shall be produced in Word (latest release).

The specifications shall be of sufficient detail so that when used with associated drawings, and with any specifications and standards incorporated by reference, construction proposals can be sought by the Government on a full and open competitive basis. The technical specifications shall precisely state what is expected from the construction contractor in clear and concise statements. All materials shall be described, where possible, by reference to standards generally known to the industry. The standards include nationally recognized industry, technical society, Federal, or military specifications and standards. Each cited specification or standard shall include its publication date. The standard which best represents no more and no less than the Government's minimum needs shall be selected for incorporation into the specification.

The 100-percent submittal for final designs shall include an electronic copy.

- C.** Cost estimates shall be accurate and realistic, reflecting current prices for work and materials, and shall take into account possible labor shortages or other local conditions that may affect the cost of the work. Sufficient supporting data shall be furnished to enable the Government to determine how the estimated costs were established.

Since the cost estimates prepared by the A/E will be used for evaluating construction contractor proposals, the estimates are considered confidential material and are intended for official government use only. This material shall not be divulged to other than authorized Government

personnel. Cost estimates shall remain the sole property of the Government and shall not be made available to others for any purpose. The sale or provision of quantity takeoffs and associated costs by the A/E to prospective bidders is prohibited. Standard units of measure, common to the trade involved, are to be used. The accuracy of the quantity takeoff shall be carefully verified. A reasonable allowance for waste shall be included where applicable. The estimate shall cover all work to be performed by the construction contractors, including handling and installation of any Government- furnished equipment. Unit prices shall be to the nearest cent and extensions to the nearest dollar.

Cost estimates shall be prepared in two parts:

1. The Estimate Summary Sheet; and
2. Supporting data consisting of quantity takeoff worksheets, indexed compilation of quotations from manufacturers and suppliers, and data to support the A/E's unit prices.

D. Catalog Cuts. Electronic copy furnished for all specified equipment.

E. Design Summary. The A/E shall prepare an electronic copy of the draft and final design summaries, as required by the task orders. The design summary shall include chapters on designer's operating criteria and construction considerations. Design data shall be included in an appendix of the design summary. The design summary shall be included as an integral part of the reports produced for the preliminary design review, progress review, draft final, and final design.

F. Design calculations shall be included in the design summary and shall consist of all functional and engineering criteria, design information, and calculations to support the design drawings and specifications. The design calculations shall be represented in a clear and legible form incorporating a title page and a table of contents. Pages shall be numbered consecutively and identified in the table of contents. Cross-referencing shall be clear. All sources of information, formula, tables, and references shall be identified. Assumptions and conclusion shall be explained.

No portion of any design analysis shall be both computed and checked by the same person.

G. Other reports and deliverables and/or changes to the above requirements will be specifically detailed in each individual task order.